Ethan Deal

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Full-Stack Software Engineer | React, TypeScript, Python, FastAPI, Node.js, RAG Systems

EDUCATION

Duke University & Duke Kunshan University | Dual Degree — **May 2026 | Kunshan 昆山, CN**

B.S. Computation and Design, Computer Science

- GPA: 3.6/4.0; Awarded Merit Scholarship (60% Tuition)

EXPERIENCE

AllEmber Systems | Founding Full-Stack Engineer — **Feb 2026 - Present | Remote**

- Co-founded an AI consultancy, building an agent-agnostic platform giving teams AI tools that act on their own company knowledge and workflows. Piloted across multiple organizations, supporting **50+ users**.
- Developed **MCP** server integrations exposing custom tools and multi-source connections to harnesses like Claude and Copilot.
- Built a provider-agnostic ingestion pipeline on **Cloud Run Jobs** that syncs OAuth-connected sources into Qdrant for hybrid retrieval, with fail-closed access control, parallelized processing and durable incremental updates.
- Designed secure multi-tenant infrastructure across **Next.js, FastAPI, PostgreSQL/Supabase, and Nango**, isolating credentials and org data to preserve tenant boundaries across services.
- Established **CI/CD** and release-gated deployment controls across service repositories, including tests, Cloud Run deployments, and PostgreSQL migrations.

BRIYA | Full-Stack Engineer — **July - Aug 2025 | Washington, DC**

Previous role: IT Specialist — **May - Aug 2024**

- Built and deployed a **MERN** stack room booking system supporting **5 campuses** and **100+ faculty**, achieving **zero downtime** since launch.
- Developed Node.js/MongoDB REST APIs for booking CRUD operations, recurring events, and conflict detection.
- Designed a React interface for desktop and mobile with multiple calendar views, drag-and-drop editing, and intuitive booking workflows.
- Deployed frontend and backend with GitHub Pages, Linode, SSL, Nginx reverse proxying, PM2, and DNS configuration.
- Previously enrolled and configured 1000+ Chromebooks, performed disk wipes, and supported device troubleshooting.

DUKE UNIVERSITY | Data Automation Engineer — **May - July 2025 | Durham, NC**

- Built a Python/Pandas batch processing script to clean and segment **10,000+ mailing records**, exporting ready-to-use CSV reports.
- Developed a Python ETL tool to consolidate multi-event datasets into venue-based Excel reports for cross-event analysis.
- Automated Qualtrics workflows to centralize large-scale survey responses into standardized spreadsheets.

VINTORY | Generative AI Design Specialist — **May - Sep 2023 | Remote**

- Built a Midjourney prompt library and ChatGPT workflow to generate structured prompt arrays for branded design production.
- Created **30+** postcards and **100+** templates/landing pages for vacation rental management clients, using AI-assisted design workflows.

PROJECTS

RAG Fact-Checking Extension — **July 2025 - March 2026**

- Designed a Chrome extension in React with a Node.js/Express backend to analyze highlighted claims in under **5 seconds**.
- Architected a **RAG** pipeline using **LangChain**, ChromaDB, and a locally hosted MiniLM embedding model.
- Benchmarked single-step agentic vs. deterministic two-step retrieval architectures, evaluating latency, retrieval accuracy, context leakage, and susceptibility to prompt-injected misinformation.

FlightOps Database Design — **Oct 2024 – Jan 2025**

- Designed a MySQL airline operations database with 10+ interrelated tables, SQL views, stored procedures, and CRUD workflows for booking, routing, assignment, and maintenance queries.
- Built a Playwright scraping pipeline to query live flight records and compiled datasets of **10,000+** airports, **7,000** aircraft, and **600** airlines.

SKILLS

Languages: Python, TypeScript, SQL, HTML/CSS, Java
Fullstack: React, Next.js | Node.js, Express, FastAPI | PostgreSQL, MongoDB, MySQL
AI & Retrieval: LangGraph, ChromaDB, Qdrant, Cohere, LiteLLM
Cloud & Tools: Docker, Git, Supabase, Google Cloud, Linode

ADDITIONAL INFORMATION

Languages: English (native), Mandarin (intermediate), Spanish (beginner)
Interests: Ultimate frisbee, digital painting, travel photography, piano, guitar, soccer